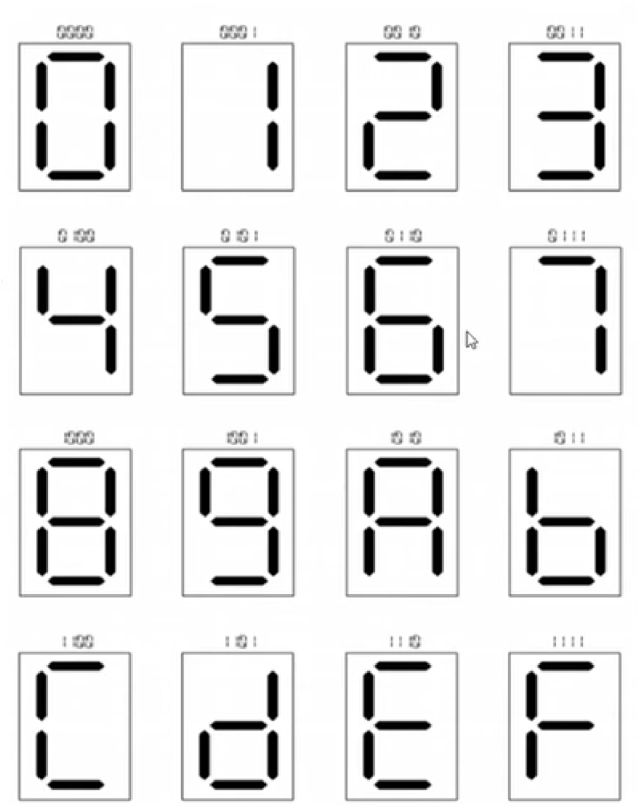
# Lab 4: BCD decoder

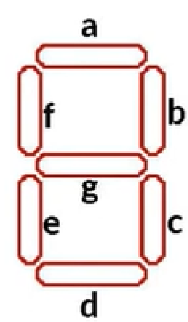
## Goals

- Design a **combinational circuit** with multiple outputs.
- Explore different methods for **digital circuit design**.

## Introduction

A **seven-segment display** is a device that uses **7 LEDs** arranged in the shape of the number 8. The individual segments are labeled **a, b, c, d, e, f, and g**, and can be turned on or off to display digits and certain characters. This type of display is widely used in **digital clocks, calculators,** and other electronic devices.






## Lab description

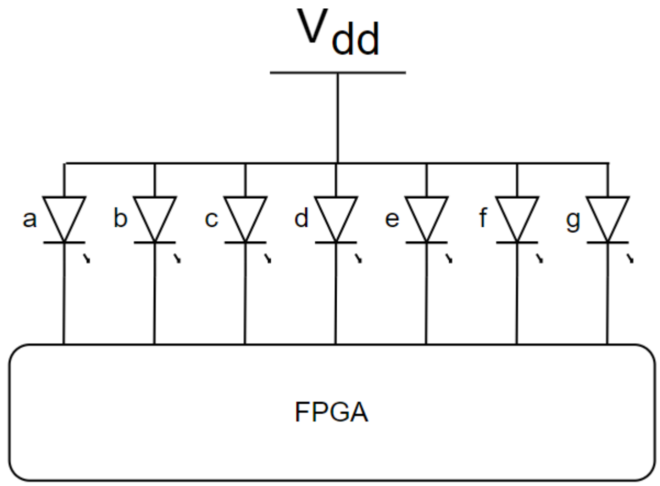
In this lab, you will design a circuit capable of displaying numbers **0 to 9** on a seven-segment display using the following methods:

- Using **logic gates** (Karnaugh maps are recommended).
- Using the **decoder method**.
- Using **LUTs (Look-Up Tables)**.

This will help analyze the **advantages and disadvantages** of each method.

**Hardware considerations**

The **seven-segment displays** on the **DE2-115 FPGA board** are **common anode**, meaning the LEDs are controlled using **logic 0 (LOW)** signals.




**Step 1**

Create the truth table.

**Step 2**

Derive the equations for each segment (LED) for **logic gate implementation**.

**Step 3**

Obtain the sum of minterms equations for each segment to implement the **decoder method**.

**Step 4**

Design the circuit using **LUTs**.

**Step 5**

Implement the circuit on the FPGA.

# Submission

**Implementation description**

Generate a document providing a detailed description of the implementation. This should include an explanation of design decisions, challenges encountered, and how they were overcome. Include diagrams, schematics, tables, or equations used in the design development.

**Video:**

Upload a short video to the Teams group, recording the operation of the implemented circuit on the FPGA and providing a brief description of its functionality.